Organisms and their environment

Mark scheme and worked solutions for 30 questions: 15 multiple choice, 15 structured.

1 · Part 1

Core · Energy flow · [1]

Answer: B

Explanation

Sunlight supplies the principal energy input, captured by producers in photosynthesis. Mineral ions and carbon dioxide supply raw materials, not energy, and heat released by decomposers is energy leaving the system, not entering it.

2 · Part 1

Core · Food chains and food webs · [1]

Answer: D

Explanation

A producer makes its own organic nutrients, usually by photosynthesis, so grass is the producer. Rabbits and hawks are consumers; the fungus is a decomposer.

3 · Part 1

Core · Food chains and food webs · [1]

Answer: A

Explanation

The grasshopper is a primary consumer feeding on the producer. The lizard eats that primary consumer and is therefore secondary.

4 · Part 1

Core · Food chains and food webs · [1]

Answer: C

Explanation

Decomposers obtain energy from dead or waste organic material. Photosynthesis identifies producers, grazing identifies herbivores, and absorbing mineral ions does not supply energy.

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5 · Part 1 Core Food chains and food webs [1]

Answer: D

Explanation

Carnivores obtain energy by eating animals. Herbivores eat plants; producers make their own organic nutrients.

6 · Part 1 Core Food chains and food webs [1]

Answer: B

Explanation

A food web links several chains and can show multiple food sources. It includes producers and does not imply energy recycling.

7 · Part 1 Core Food chains and food webs [1]

Answer: C

Explanation

Removing predators reduces mortality of their small-fish prey, allowing those fish to increase. Predation on small fish falls rather than rises, and more small fish would eat more water fleas, so water fleas would decrease, not increase.

8 · Part 1 Core Food chains and food webs [1]

Answer: A

Explanation

A pyramid of biomass takes the amount of living material at each level into account, so it is not inverted. A pyramid of numbers counts a tree and a small herbivore as one each; biomass does not assume equal mass, is not an energy-flow rate per year and does include the producer level.

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9 · Part 1 Supplement Food chains and food webs [1]

Answer: B

Explanation

Respiration releases energy from food and much of it is eventually transferred to the environment as heat. Digestion breaks food into small molecules without transferring energy out of the body, growth stores chemical energy in new tissue, and consumers do not photosynthesise.

10 · Part 1 Supplement Food chains and food webs [1]

Answer: C

Explanation

Energy transfer is inefficient, leaving less available at higher levels. This limits the populations that a higher trophic level can support.

11 · Part 1 Supplement Food chains and food webs [1]

Answer: D

Explanation

Cattle use energy in respiration and lose energy in faeces and urine, so they store only a small fraction of the crop energy. Direct consumption avoids this additional transfer step. Humans digest plant material less well than cattle, and plants do respire.

12 · Part 1 Supplement Nutrient cycles [1]

Answer: A

Explanation

Nitrogen fixation converts nitrogen gas into nitrogen compounds such as ammonium ions. Denitrification does the reverse, releasing nitrogen gas; nitrification converts ammonium ions to nitrate ions, and decomposition converts protein to ammonium ions. Neither of the last two starts from nitrogen gas.

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13 · Part 1 Core Populations [1]

Answer: C

Explanation

A population contains organisms of one species in the same area at the same time. A community includes different species.

14 · Part 1 Core Populations [1]

Answer: A

Explanation

A community includes all the populations of different species. One species is a population; physical conditions alone are not a community.

15 · Part 1 Core Populations [1]

Answer: B

Explanation

Disease increases mortality or reduces reproduction and therefore limits population growth. It does not make survivors reproduce faster, it is not restricted to old deer, and a disease of deer does not kill their predators.

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16 · Part 1 Core Energy flow [4]

Mark scheme

(a) **[2]**

- Green plants absorb light energy from the Sun. (1)
- Photosynthesis stores energy as chemical energy in organic nutrients. (1)

(b) **[2]**

- Feeding transfers chemical energy between organisms. (1)
- Energy is eventually transferred to the environment, for example as heat from respiration. (1)

Worked solution

(a) Green plants absorb light energy from the Sun. Photosynthesis stores energy as chemical energy in organic nutrients.

(b) Feeding transfers chemical energy between organisms. Energy is eventually transferred to the environment, for example as heat from respiration.

17 · Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- pondweed → snail → fish → heron, with all four organisms in the correct order (1)
- arrows pointing from the organism eaten to the organism that eats it (1)

> **REJECT** arrows pointing towards the producer

(b) **[2]**

- Pondweed is the producer. (1)
- The heron is the tertiary consumer. (1)

Worked solution

(a) pondweed → snail → fish → heron. The chain starts with the producer and each arrow points from the organism being eaten to the organism that eats it, showing the direction of energy transfer.

(b) Pondweed is the producer. The heron is the tertiary consumer.

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18 · Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- Owls have less available food. (1)
- Owl survival or reproduction may decrease, reducing their population. (1)

(b) **[2]**

- Foxes can also eat rabbits. (1)
- This alternative food source reduces their dependence on voles. (1)

Worked solution

(a) Owls have less available food. Owl survival or reproduction may decrease, reducing their population.

(b) Foxes can also eat rabbits. This alternative food source reduces their dependence on voles.

19 · Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- It obtains energy by feeding on other organisms. (1)
- Those organisms are plants, so it is a herbivore. (1)

(b) **[2]**

- It obtains energy by eating animals. (1)
- It is a secondary consumer in the leaf → hare → wildcat chain. (1)

Worked solution

(a) It obtains energy by feeding on other organisms. Those organisms are plants, so it is a herbivore.

(b) It obtains energy by eating animals. It is a secondary consumer in the leaf → hare → wildcat chain.

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20 · Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- They obtain energy from dead organic material. (1)
- They are decomposers. (1)

(b) **[2]**

- Plants make organic nutrients through photosynthesis. (1)
- They use light energy rather than obtaining all energy by consuming dead material. (1)

Worked solution

(a) They obtain energy from dead organic material. They are decomposers.

(b) Plants make organic nutrients through photosynthesis. They use light energy rather than obtaining all energy by consuming dead material.

21 · Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- Counting gives each tree and each insect equal weight. (1)
- It ignores the much greater amount of living material in one tree. (1)

ALLOW a pyramid of numbers would be inverted / top-heavy, suggesting too few producers

(b) **[2]**

- Biomass accounts for differences in organism size. (1)
- It better represents the amount of material available at each trophic level. (1)

Worked solution

(a) Counting gives each tree and each insect equal weight. It ignores the much greater amount of living material in one tree.

(b) Biomass accounts for differences in organism size. It better represents the amount of material available at each trophic level.

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22 · Part 1 Supplement Food chains and food webs [4]

Mark scheme

(a) **[2]**

- It represents the energy available to support organisms at each level. (1)
- It measures transfer over time rather than just a snapshot of stored biomass. (1)

(b) **[2]**

- Some energy is used in respiration and transferred as heat. (1)
- Some material is not eaten or is not digested and absorbed. (1)

Worked solution

(a) It represents the energy available to support organisms at each level. It measures transfer over time rather than just a snapshot of stored biomass.

(b) Some energy is used in respiration and transferred as heat. Some material is not eaten or is not digested and absorbed.

23 · Part 1 Supplement Food chains and food webs [4]

Mark scheme

(a) **[2]**

- 2700 ÷ 18 000 (× 100) (1)
- = 15% (1)

ALLOW 2 marks for correct answer with no working

(b) **[2]**

- Not all plant material is eaten or digested. (1)
- Herbivores use absorbed energy in respiration, transferring energy to the environment. (1)

Worked solution

(a) 2700 divided by 18 000 gives 0.15. Multiplying by 100 gives 15%.

(b) Not all plant material is eaten or digested. Herbivores use absorbed energy in respiration, transferring energy to the environment.

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24 · Part 1 Supplement Food chains and food webs [4]

Mark scheme

(a) **[2]**

- It has fewer trophic transfers. (1)
- It avoids energy used by livestock in respiration and lost in uneaten or waste material. (1)

(b) **[2]**

- Each transfer passes on only a fraction of the available energy. (1)
- Eventually too little energy remains to support a further substantial consumer population. (1)

Worked solution

(a) It has fewer trophic transfers. It avoids energy used by livestock in respiration and lost in uneaten or waste material.

(b) Each transfer passes on only a fraction of the available energy. Eventually too little energy remains to support a further substantial consumer population.

25 · Part 1 Core Food chains and food webs [4]

Mark scheme

(a) **[2]**

- The introduced insect reduces the birds' food supply. (1)
- Increased competition may reduce bird survival and reproduction. (1)

(b) **[2]**

- Fewer birds provide less food for their predator. (1)
- The predator population may then decrease or switch to other prey. (1)

Worked solution

(a) The introduced insect reduces the birds' food supply. Increased competition may reduce bird survival and reproduction.

(b) Fewer birds provide less food for their predator. The predator population may then decrease or switch to other prey.

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26 · Part 1 Core Nutrient cycles [4]

Mark scheme

(a) **[2]**

- Plants absorb carbon dioxide during photosynthesis. (1)
- Herbivores obtain carbon compounds by feeding on plants. (1)

(b) **[2]**

- Respiration by plants, animals or decomposers. (1)
- Combustion of wood or fossil fuels. (1)

ALLOW decomposition (by decomposers) as one of the two processes

Worked solution

(a) Plants absorb carbon dioxide during photosynthesis. Herbivores obtain carbon compounds by feeding on plants.

(b) Respiration by plants, animals or decomposers. Combustion of wood or fossil fuels.

27 · Part 1 Supplement Nutrient cycles [4]

Mark scheme

(a) **[2]**

- Decomposers convert nitrogen in proteins to ammonium ions. (1)
- Nitrifying bacteria convert ammonium compounds to nitrate ions that plants can absorb. (1)

(b) **[2]**

- Nitrogen fixation converts nitrogen gas to nitrogen compounds. (1)
- Denitrification converts nitrate ions to nitrogen gas, returning it to the atmosphere. (1)

Worked solution

(a) Decomposers convert nitrogen in proteins to ammonium ions. Nitrifying bacteria convert ammonium compounds to nitrate ions that plants can absorb.

(b) Nitrogen fixation converts nitrogen gas to nitrogen compounds. Denitrification converts nitrate ions to nitrogen gas, returning it to the atmosphere.

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28 · Part 1 Core Populations [4]

Mark scheme

(a) **[2]**

- The reed population is the organisms of that one species in the area at the same time. (1)
- The community contains all populations of different species in the wetland. (1)

(b) **[2]**

- It includes a community and its physical environment. (1)
- The organisms and the environment interact with one another. (1)

Worked solution

(a) The reed population is the organisms of that one species in the area at the same time. The community contains all populations of different species in the wetland.

(b) It includes a community and its physical environment. The organisms and the environment interact with one another.

29 · Part 1 Core Populations [4]

Mark scheme

(a) **[2]**

- Rabbits compete more for limited food. (1)
- Less food can reduce survival or reproduction. (1)

(b) **[2]**

- More foxes can kill more rabbits. (1)
- This increases mortality and reduces net population growth. (1)

Worked solution

(a) Rabbits compete more for limited food. Less food can reduce survival or reproduction.

(b) More foxes can kill more rabbits. This increases mortality and reduces net population growth.

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30 · Part 1 Supplement Populations [4]

Mark scheme

(a) **[2]**

- Initially the bacteria adapt to the conditions and there are few reproducing cells. (1)
- Later many bacteria reproduce while nutrients and other resources are plentiful. (1)

(b) **[2]**

- Nutrients become limiting and wastes accumulate. (1)
- Birth and death rates become approximately equal in the stationary phase. (1)

Worked solution

(a) Initially the bacteria adapt to the conditions and there are few reproducing cells. Later many bacteria reproduce while nutrients and other resources are plentiful.

(b) Nutrients become limiting and wastes accumulate. Birth and death rates become approximately equal in the stationary phase.

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